Reader Supplement to

Generic Identifiability and Directed Containment for Strongly Tree-Child Level-2 Networks under the Kimura Two-Parameter Model: The Principal Positive Domain and Strict Continuous Time

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Abstract

This supplement is a reader-oriented map from the mathematical proof to the exact finite certificate package. It separates hand proofs from the bounded computer-assisted residue, defines the primitive completion grammar and raw relation object, records all censuses and restoration/probe invariants, gives representative certificate schemas and the complete weak-class witness, and lists the minimal artifacts and commands needed to replay each theorem layer. The finite principal-domain theorem and classification universe remain unchanged; the qualification-evidence lock is resealed when verifier-facing tests or their provenance bindings are repaired. The article and this supplement are separately versioned submission sources.

Contents

1 How to read the proof

The main article proves a global theorem from three qualitatively different types of input.

P1. Analytic and algebraic proofs. These include the physical K2P domain, subdivision and rerooting, pointwise quartet and whole-map $\mathcal{T}_i$ signs, the two-sector bridge fibre, physical local products, marginal submersions, localization, genericity, continuous-time gluing, and the weak-class construction.

P2. Topology-only finite reduction. The cycle/theta primitive theorem and minimum-repair grammar are graph-theoretic. They reduce an arbitrary strong level-2 factor to a finite rigid support plus ordered labelled words on finitely many directed segments.

P3. Exact bounded computation. Raw four- and five-port directions, rank exclusions, direct polynomial separators, restoration children, and coherent probes are exhausted by exact graph and rational/symbolic arithmetic. This layer has independent regeneration and mutations, but is not presented as a complete short hand classification.

The logical dependency is

$$\begin{gathered}\text{pointwise decorated tree of blobs}\\ \downarrow\\ \text{complete two-sector bridge quotient and physical product chart}\\ \downarrow\\ \text{finite localization to one complete ported factor}\\ \downarrow\\ \text{bounded raw classification} + \text{fixed-full restoration}\\ \downarrow\\ \text{one-/two-port reconstruction of arbitrary words}\\ \downarrow\\ \text{global necessity} \quad + \quad \text{ordinary-triangle common germ}\end{gathered}$$

The quartet and $\mathcal{T}_i$ inequalities are pointwise. The polynomial and rank certificates are generic non-containment statements. Restoration applies only inside one assumed fixed full relation. Keeping these quantifiers separate is essential.

The independent replay boundary is also explicit. The package regenerates the primitive universes, checks exact graph relations and symbolic certificates, and subjects complete ledgers to verifier-facing mutations. It does not claim an additional all-family orbit partition built without either the submitted primitive atlas or canonicalizer, nor a second symbolic engine that independently re-expands every higher-degree polynomial body. Those would be stronger independent research audits, not premises of the theorem or of the finite replay reported here.

2 Convention crosswalk

All graphs use the following fixed convention.

- Rooted presentations are binary directed acyclic graphs; the root is the lowest stable ancestor.
- Only arcs entering reticulations retain arrowheads.
- The root is deleted once and its two incident arcs merge into one mixed edge. No later parallel-edge or degree-two cleanup is part of formation of the full standard semi-directed graph.
- Restriction/display operations may subsequently prune dead ends and suppress unlabelled bivalent vertices; they do not change the admissible-rooting universe of the original graph.
- "Strongly tree-child" quantifies over every admissible rooting; "weakly tree-child" requires at least one tree-child rooting.
- Isomorphisms preserve labels, ordinary edges, arrowheads, and vertex roles. Ordinary triangle equivalence forgets only the reticulation choice on one three-cycle and keeps its three external arms fixed.

The K2P character order is $(0, C, G, T)$, with equal sector $\{C, T\}$ and singleton sector $\{G\}$. Every edge has Fourier vector $(1, s, g, s)$ in

$$\mathcal{D}_+ = \{0 < s < 1,\ 0 < g < 1,\ g > 2s - 1\}.$$

The strict continuous-time subdomain is

$$\mathcal{D}_{\mathrm{CT}} = \{0 < s < 1,\ s^2 < g < 1\}.$$

No mixed-sign stochastic component is part of the theorem.

3 Primitive graph grammar

3.1 Core segments and minimum repairs

After removing external port arms and suppressing ordinary non-event degree-two vertices, a nontrivial strong level-2 blob is a cycle or one of four directed theta event placements. Segment indices in the last column below refer to the left-to-right segment order.

core	directed segments	minimum repair sets
cycle	$S \to X, S \to X$	$\{0\}, \{1\}$
θ_0	$S \to U, S \to V, U \to X, V \to X, U \to V$	$\{2,3\}, \{3,4\}$
θ_1	$S \to U, S \to X, V \to X, U \to V, U \to V$	$\{2,3\}, \{2,4\}$
θ_2	$S \to U, S \to V, U \to X_0, V \to X_0, U \to X_1, V \to X_1$	$\{2,3\}, \{2,5\}, \{3,4\}, \{4,5\}$
θ_3	$S \to U, S \to X_0, V \to X_0, U \to X_1, V \to X_1, U \to V$	$\{2\}, \{4\}$

An occupied segment contains at least one port-bearing subdivision. The minimum repairs are the minimal occupancy sets satisfying the no-omnian and simple-graph requirements. A complete factor additionally retains one child role below every path-sink reticulation. Here "occupied" includes a named zero-character completion role inserted by the compiler; it need not be a selected physical port. Arbitrary physical factors arise by replacing each selected segment word by an ordered word of distinct labelled ports.

3.2 Completion grammar and the numbers 831, 1983, and 4155

For each core the compiler enumerates:

(i) every allowed subset of path-sink child roles;
(ii) every weak composition of the remaining labelled port roles among the directed segments;
(iii) for a theta target, every minimum-repair tag; for a cycle target, the single untagged target convention (the two repaired cycle supports occur on the source side);
(iv) the selected-incoming mode, in which a real incoming boundary remains a port, and the marginalized-incoming mode, in which that role has been omitted; and
(v) every physical port permutation.

The directed completion grammar yields 831 four-port target descriptors in selected-incoming mode and 1,983 in marginalized-incoming mode. Five-port targets number 1,983 and 4,155 in the corresponding modes. These are repair-tagged descriptors, not necessarily distinct untagged mixed graphs. The compiler first weakly distributes the selected physical labels. For a chosen theta repair it then inserts one deterministically named zero-character leaf on each required segment whose physical word is empty, and inserts nothing extra on an already occupied required segment. The repair tag stays in the raw identity even when another repair witness yields the same final mixed graph. Unselected sink roles and a marginalized incoming role receive their own deterministic dummies. Consequently these roles consume no selected labels and the stars-and-bars count in the article is unchanged. The numbers are taken before source multiplication and port permutations. Thus

$$6(831 + 1983)4! = 405{,}216, \qquad 4(1983 + 4155)5! = 2{,}946{,}240.$$

3.3 Raw relation schema

A raw directional relation is not an unlabelled topology hash. Its immutable identity includes:

- source primitive and exact repaired support;
- target primitive, completion descriptor, and selected graph;
- source and target incoming roles;
- physical port permutation and port match;
- omitted source and target roles;
- source-to-target direction;
- boundary incidence transport; and
- exact graph fingerprints.

Only after every raw ID is generated exactly once are presentations grouped into canonical terminal or restoration classes.

4 Certificate decision tree

The classifier uses proof types in a fixed precedence order.

type	mathematical implication	exact stored evidence
Displayed quartet	physical images are disjoint pointwise	distinct complete switching-split sets, selected witness split, and the appropriate leaf-permuted F or G polynomial.
Whole-map $\mathcal{T}_i$	physical images are disjoint pointwise	coordinate triple and orientation, exact whole-map pullbacks, and a strict Bernstein/sign certificate for the factorized pullback.
Directed rank	no full-dimensional source germ enters the target	nonzero exact source minor, target lower minor, symbolic target rank upper certificate, and strict source-rank > target-rank comparison.
Polynomial separator	generic source-to-target containment is impossible	sparse integer polynomial $P(q)$, coefficientwise proof $P \circ \Phi_T = 0$, nonzero source pullback, and strict rational source witness.
Restoration parent	deferred fixed-full obligation	parent relation, each actual physical child attachment, exact parent–child transport restriction, and a terminal certificate at every leaf.
Isomorphism	physical parameter transport exists	exact labelled mixed-incidence isomorphism and parameter/port transport.
Ordinary triangle	one common full-dimensional regular germ exists	stored triangle edge set and coherent transport, combined with the exact rank-nine analytic construction in the article.

The target rank upper certificate is symbolic; a one-point Jacobian rank is never used as an upper bound. Polynomial witnesses are checked on the graph- generated K2P maps rather than accepted by a topology identifier.

5 Finite census and current compression boundary

5.1 Authoritative raw partitions

category	four-port theta	five-port θ_2
displayed quartet	360,408	2,942,592
whole-map $\mathcal{T}_i$	16,974	2,528
directed exact rank	23,822	800
direct terminal / quadratic	1,472	240
restoration member / isomorphism	2,540	80
total	405,216	2,946,240

The 1,472 four-port terminal presentations form 934 classes:

$$839 \text{ quadratics} + 36 \text{ direct polynomials} + 4 \text{ hard bindings} + 20 \text{ isomorphisms} + 35 \text{ triangles}.$$

The 36 direct polynomials have three certified source-family tags: 22 θ_0 quintic port-orbit obstructions, 12 lower-theta quartics, and two θ_3 cubics. The 839 quadratic classes have eight literal certificate bodies. These repetitions make further theorem-template compression plausible, but equality of certificate bodies is not by itself a proof that all graph transports belong to one legal orbit.

5.2 Audited proof-compression table

The bounded compression pass has status `PC-PARTIAL`. It changes no theorem statement and discards no exact ledger. In particular, literal equality of polynomial bodies is not treated as a graph orbit, and the 297 restoration fingerprints are descriptive archetypes rather than a proved transport quotient.

layer	frozen units	compact units	exact residue retained
four-port quadratics	839	8	every coordinate transport
five-port θ_2 quadratics	96	4	every coordinate transport
three-port cycle quadratics	54	6	every coordinate transport
restoration quadratics	6	5	every coordinate transport
direct-36 high degree	36	27	all directional bindings; 3 certified proposition families, 5 stored base formulas
rank upper certificates	4,379	3,515+75	universal mechanism plus 75 exceptional representatives covering 864 descriptors
restoration parents	997	297	997 assignments, 2,540 roots, 36,824 edges, and 16 algebra-transport classes

The direct-36 row must not be read as three literal polynomials. The quintic, quartic, and cubic proposition families have 27 direction-specific bodies across the 36 records. The quadratic rows group only literal coordinate bodies inside their original coordinate convention or transports already certified by a frozen orbit package.

layer	uniform statement	load-bearing finite premise
primitive completions	one binomial-sum formula for all selected/marginalized cases	all labelled directed completion records; totals 405,216, 2,946,240, and 13,440
restoration	four terminal mechanisms and one depth-two continuation mechanism	the exact 997-parent assignment and every physical parent–child transport
probe reconstruction	one-/two-port coherence implies arbitrary attachment-word recovery	176 anchors, 29,964 one-port rows, 544,571 two-port rows, 67,741 transports, and 4,379 restrictions

The machine-readable theorem-to-artifact crosswalk is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`. It records, for each theorem layer, exact file hashes and declared schemas, producer, replay, mutation, and environment bindings. The original frozen principal-domain computational-evidence lock does not itself contain an end-to-end timing record. For C01–C10 and C12–C13, the theorem-layer crosswalk leaves end-to-end runtimes null, preserving locked component observations without summing them. Separately, its global C11 entry and the submission bundle bind a detached clean-checkout replay of the qualified lock. The exact candidate commit, layer count, wall and internal times, memory measurements, report hash, and telemetry hash are recorded only in the externally bound replay artifacts and generated theorem-to-artifact crosswalk; they are deliberately not compiled into this source. No byte-bound end-to-end quick-suite runtime is claimed.

5.3 Restoration forests

The corrected four-port restoration forest has 997 canonical parents and 2,540 physical member roots. Its first layer is

$$\begin{array}{lr} \text{quartet separator} & 35,758 \\ \text{whole-map } \mathcal{T}_i & 606 \\ \text{quadratic} & 148 \\ \text{transported quartic} & 24 \\ \text{continuation} & 32 \end{array} \qquad \text{(36,568 first children).}$$

The 32 continuations have 256 second children: 248 quartet and eight $\mathcal{T}_i$. Hence there are 36,824 edges, 36,792 exact separator leaves, maximum depth two, no cycle, no missing child, and no unresolved record. Every child transport restricts to its parent's boundary transport.

The five-port θ_2 dummy forest has 56 roots and 864 one-parent descendants. Its 832 leaves are 760 quartet exclusions and 72 labelled isomorphisms. The cycle layer has 5,964 restoration roots and 536,364 physical completions, of which 535,920 are quartet-separated, 300 are $\mathcal{T}_i$-separated, 132 have exact quadratics, and 12 are isomorphic.

The logical implication is always fixed-full. An assumed full relation fixes the omitted physical label, source edge, target edge, role, and transport. That actual completion is one child in the forest. Source marginal openness then turns its separator into a contradiction. No abstract parent relation is lifted, and no target deletion map is inverted.

6 Coherent probes and arbitrary subdivision words

The equality-anchor input contains 176 physical relations:

origin	ports	records
four-port direct terminals	4	26
restored four-port terminals	5	17
θ_2 direct anchors	5	24
θ_2 one-role restorations	6	40
θ_2 two-role paths	7	32
cycle direct anchors	3	24
cycle restored anchors	4	12
ordinary tree identity	3	1
total		176

There are 143 isomorphism and 33 triangle anchors.

If a rooted binary factor has k boundary leaves and r reticulations, degree counting gives $t = k + r - 2$ nonroot tree vertices and $2k + 3r - 2$ rooted arcs. Root suppression merges two arcs, leaving exactly

$$2k + 3r - 3$$

physical attachment sites. The probe input contains every site: the root-suppressed incoming segment, every other pendant arm, both incoming edges at every reticulation, and every remaining unheaded core edge. The two artificial root halves are identified only after exact semi-directed isomorphism.

Across all anchors there are 2,206 source sites and 2,206 target sites. The first-probe universe is the per-anchor Cartesian product and contains 29,964 directions. Its census is

$$27,758 \text{ quartet} + 99\ \mathcal{T}_i + 1,915 \text{ isomorphism} + 192 \text{ triangle}.$$

The 2,107 equality survivors generate every two-port parent exactly once. The second-probe universe has 544,571 directions:

$$511,266 \text{ quartet} + 576\ \mathcal{T}_i + 30,969 \text{ isomorphism} + 1,760 \text{ triangle}.$$

One-port restrictions identify the directed core segment containing each omitted boundary. Two-port restrictions determine the order of every pair on one segment. Since the comparisons come from a genuine word, they are transitive and recover a unique total order. The exact child transports agree on overlaps and restrict to their parents. All 33 triangle anchors transport the same inherited ordinary triangle; probes create no new triangle. This is the finite mechanism that extends the bounded theorem from rigid supports to arbitrary complete labelled subdivision words.

7 Representative exact calculations

7.1 Tree–sunlet factorization

For the three-sunlet orientation at leaf three,

$$q_{xyz} = a_x b_y c_z\{\delta f_y d_z + (1-\delta) f_x e_z\}, \qquad x + y + z = 0.$$

With $X_h = q_{hh0}$, $Y_h = q_{h0h}$, $Z_h = q_{0hh}$, and $V = q_{CTG}$, exact substitution gives

$$V^2 X_g - X_s^2 Y_g Z_g = -a_s^2 b_s^2 a_g b_g c_g^2 f_s^2 \delta(1-\delta) d_g e_g (1-f_g)^2.$$

This identity is used on whole semi-directed maps. A rooted restriction type is not used as a topology oracle.

7.2 Bridge normalizer

For an unmarked degree-d component, the pair-anchor rows $(12), (13), (23), (14), \ldots, (1d)$ have rank d and leading determinant -2. If r_{ij} are positive anchor ratios, the incidence scales are

$$a_1 = \sqrt{r_{12}r_{13}/r_{23}}, \quad a_2 = \sqrt{r_{12}r_{23}/r_{13}}, \quad a_3 = \sqrt{r_{13}r_{23}/r_{12}}, \quad a_k = r_{1k}/a_1.$$

The formula is applied independently to the $\{C, T\}$ and G sectors.

7.3 Certificate schema for a direct polynomial

A sparse direct certificate stores

$$P(q) = \sum_m c_m \prod_{j \in m} q_j,$$

with integer coefficients and explicit Fourier-coordinate tuples. Its replay regenerates source and target polynomial maps from the two graph encodings, forms $P \circ \Phi_S$ and $P \circ \Phi_T$ over $\mathbb{Z}$, checks $P \circ \Phi_T = 0$ coefficientwise, checks a nonzero source polynomial hash, and evaluates it at a rational point satisfying every strict K2P and inheritance inequality. Degrees two, three, four, and five are stored and mutated separately so that certificate reassignment is rejected.

8 Printed exact-certificate appendix

This appendix exposes the formulas already derived by the first compression cycle; it is not a second compression claim. The 23 quadratic templates identify literal coefficient, coordinate-pair, and multidegree bodies only. Equality of displayed formulas across layers does not license any graph, port, direction, or coordinate transport. In particular, the repeated literal bodies in different layers remain different graph-derived certificates.

8.1 Coordinate dictionaries

Write the character group in the integer order $0, C, G, T$. A coordinate index is obtained by listing the zero-sum tuples, replacing each tuple by the lexicographically smaller of it and its global C/T swap, deleting duplicates, and sorting. Thus $q_i = q_{h_1 \cdots h_k}$ for the tuple printed below. All 36 four-port coordinates occur in the printed quadratic or higher-degree formulas:

index	tuple	index	tuple	index	tuple	index	tuple
q_0	0000	q_1	$00CC$	q_2	$00GG$	q_3	$0C0C$
q_4	$0CC0$	q_5	$0CGT$	q_6	$0CTG$	q_7	$0G0G$
q_8	$0GCT$	q_9	$0GG0$	q_{10}	$C00C$	q_{11}	$C0C0$
q_{12}	$C0GT$	q_{13}	$C0TG$	q_{14}	$CC00$	q_{15}	$CCCC$
q_{16}	$CCGG$	q_{17}	$CCTT$	q_{18}	$CG0T$	q_{19}	$CGCG$
q_{20}	$CGGC$	q_{21}	$CGT0$	q_{22}	$CT0G$	q_{23}	$CTCT$
q_{24}	$CTG0$	q_{25}	$CTTC$	q_{26}	$G00G$	q_{27}	$G0CT$
q_{28}	$G0G0$	q_{29}	$GC0T$	q_{30}	$GCCG$	q_{31}	$GCGC$
q_{32}	$GCT0$	q_{33}	$GG00$	q_{34}	$GGCC$	q_{35}	$GGGG$

The following are exactly the 52 five-port indices occurring in the stored literal coordinate-pair blocks behind the printed theta2 and restoration quadratics. Of these, 46 have a nonzero displayed coefficient; six occur only in zero slots retained by the exact certificate-body normalization:

index	tuple	index	tuple	index	tuple	index	tuple
q_1	$000CC$	q_2	$000GG$	q_3	$00C0C$	q_4	$00CC0$
q_5	$00CGT$	q_6	$00CTG$	q_7	$00G0G$	q_8	$00GCT$
q_9	$00GG0$	q_{10}	$0C00C$	q_{18}	$0CG0T$	q_{26}	$0G00G$
q_{27}	$0G0CT$	q_{28}	$0G0G0$	q_{29}	$0GC0T$	q_{30}	$0GCCG$
q_{31}	$0GCGC$	q_{32}	$0GCT0$	q_{34}	$0GGCC$	q_{35}	$0GGGG$
q_{36}	$C000C$	q_{37}	$C00C0$	q_{40}	$C0C00$	q_{41}	$C0CCC$
q_{43}	$C0CTT$	q_{49}	$C0TCT$	q_{51}	$C0TTC$	q_{52}	$CC000$
q_{53}	$CC0CC$	q_{55}	$CC0TT$	q_{56}	$CCC0C$	q_{57}	$CCCC0$
q_{64}	$CCT0T$	q_{67}	$CCTT0$	q_{85}	$CT0CT$	q_{87}	$CT0TC$
q_{88}	$CTC0T$	q_{91}	$CTCT0$	q_{96}	$CTT0C$	q_{97}	$CTTC0$
q_{100}	$G000G$	q_{102}	$G00G0$	q_{103}	$G0C0T$	q_{105}	$G0CGC$
q_{109}	$G0GGG$	q_{110}	$GC00T$	q_{118}	$GCG0C$	q_{128}	$GG0GG$
q_{129}	$GGC0C$	q_{131}	$GGCGT$	q_{133}	$GGG0G$	q_{135}	$GGGG0$

8.2 The 23 literal quadratic bodies

The multidegree is ordered as $(s_0, g_0, s_1, g_1, \ldots)$. "Canonical" counts exact certificate classes; "raw" counts their raw directional presentations.

ID	layer	polynomial	multidegree	canonical	raw
R4Q-01	four-port	$q_4q_{20} + q_5q_{21} - q_8q_{24} - q_9q_{15} - q_9q_{17} + q_9q_{23}$	$(1,0,1,1,1,1,1,0)$	12	12
R4Q-02	four-port	$q_{10}q_{30} + q_{13}q_{29} - q_{15}q_{26} - q_{17}q_{26} - q_{22}q_{27} + q_{23}q_{26}$	$(1,1,1,0,1,0,1,1)$	52	84
R4Q-03	four-port	$q_{10}q_{20} - q_{12}q_{18}$	$(2,0,0,1,0,1,2,0)$	252	424
R4Q-04	four-port	$q_3q_{19} - q_6q_{18} - q_7q_{23} + q_8q_{22}$	$(1,0,1,1,1,0,1,1)$	24	40
R4Q-05	four-port	$q_3q_{19} + q_6q_{18} - q_7q_{23} - q_8q_{22}$	$(1,0,1,1,1,0,1,1)$	68	108
R4Q-06	four-port	$q_1q_{34} - q_8q_{27}$	$(0,1,0,1,2,0,2,0)$	292	456
R4Q-07	four-port	$q_{11}q_{31} + q_{12}q_{32} - q_{23}q_{28} - q_{24}q_{27}$	$(1,1,1,0,1,1,1,0)$	8	8
R4Q-08	four-port	$q_1q_{16} - q_2q_{17} + q_5q_{13} - q_6q_{12}$	$(1,0,1,0,1,1,1,1)$	131	220
T2Q-01	five-port theta2	$q_2q_{133} - q_7q_{128} - q_{26}q_{109} + q_{35}q_{100}$	$(0,1,0,1,0,1,0,1,0,2)$	24	60
T2Q-02	five-port theta2	$q_2q_{135} - q_9q_{128} - q_{28}q_{109} + q_{35}q_{102}$	$(0,1,0,1,0,1,0,2,0,1)$	24	60
T2Q-03	five-port theta2	$q_{36}q_{91} + q_{37}q_{64} - q_{40}q_{87} - q_{49}q_{52}$	$(2,0,1,0,1,0,1,0,1,0)$	16	40
T2Q-04	five-port theta2	$q_{36}q_{57} - q_{36}q_{67} - q_{37}q_{56} + q_{37}q_{88} + q_{40}q_{55} - q_{40}q_{87} - q_{43}q_{52} + q_{49}q_{52}$	$(2,0,1,0,1,0,1,0,1,0)$	32	80
C3Q-01	three-port cycle in four-port coordinates	$q_4q_{20} + q_5q_{21} - q_8q_{24} - q_9q_{15} - q_9q_{17} + q_9q_{23}$	$(1,0,1,1,1,1,1,0)$	2	4
C3Q-02	three-port cycle in four-port coordinates	$q_4q_{30} - q_6q_{32}$	$(0,1,2,0,2,0,0,1)$	16	40
C3Q-03	three-port cycle in four-port coordinates	$q_1q_{34} - q_8q_{27}$	$(0,1,0,1,2,0,2,0)$	16	40
C3Q-04	three-port cycle in four-port coordinates	$q_1q_{16} - q_2q_{17} - q_5q_{13} + q_6q_{12}$	$(1,0,1,0,1,1,1,1)$	2	4
C3Q-05	three-port cycle in four-port coordinates	$q_1q_{16} - q_2q_{17} + q_5q_{13} - q_6q_{12}$	$(1,0,1,0,1,1,1,1)$	2	4
C3Q-06	three-port cycle in four-port coordinates	$q_{11}q_{19} - q_{13}q_{21}$	$(2,0,0,1,2,0,0,1)$	16	40
RSQ-01	five-port restoration	$q_3q_{131} - q_5q_{129} - q_{29}q_{105} + q_{31}q_{103}$	$(0,1,0,1,2,0,0,1,2,0)$	1	1

ID	layer	polynomial	multidegree	canonical	raw
RSQ-02	five-port restoration	$q_1q_{34} - q_8q_{27}$	$(0,0,0,1,0,1,2,0,2,0)$	2	6
RSQ-03	five-port restoration	$q_4q_{30} - q_6q_{32}$	$(0,0,0,1,2,0,2,0,0,1)$	1	10
RSQ-04	five-port restoration	$q_{10}q_{118} - q_{18}q_{110}$	$(0,1,2,0,0,1,0,0,2,0)$	1	130
RSQ-05	five-port restoration	$q_3q_{131} - q_5q_{129} + q_{29}q_{105} - q_{31}q_{103}$	$(0,1,0,1,2,0,0,1,2,0)$	1	1

8.3 Five high-degree bases and their licensed transports

Only the 22 port permutations, 12 quartic coordinate transports, and two cubic record bindings printed here are licensed. No other port, source-target, inheritance, pole/sink, or C/T transport is inferred.

HDQ-01: theta0 quintic. Degree 5, multidegree $(2,1,1,1,2,1,1,1)$.

$$\begin{aligned}F_{HDQ-01} = {} & q_0q_0q_{11}q_{17}q_{35} - q_0q_0q_{11}q_{25}q_{35} - 2q_0q_0q_{17}q_{19}q_{28} + 2q_0q_0q_{19}q_{25}q_{28} \\ & - 2q_0q_1q_{11}q_{14}q_{35} + 2q_0q_1q_{13}q_{24}q_{33} + 4q_0q_1q_{14}q_{19}q_{28} - 2q_0q_1q_{21}q_{24}q_{26} \\ & + 2q_0q_2q_{10}q_{21}q_{32} + q_0q_2q_{11}q_{17}q_{33} - 2q_0q_2q_{11}q_{21}q_{29} - 2q_0q_2q_{11}q_{23}q_{33} \\ & + q_0q_2q_{11}q_{25}q_{33} - 2q_0q_2q_{14}q_{21}q_{27} + 2q_0q_4q_{10}q_{11}q_{35} - 4q_0q_4q_{10}q_{19}q_{28} \\ & - 2q_0q_4q_{12}q_{13}q_{33} + 2q_0q_4q_{12}q_{21}q_{26} - 2q_0q_5q_{11}q_{13}q_{33} + 2q_0q_5q_{11}q_{21}q_{26} \\ & + q_0q_7q_{11}q_{17}q_{28} - q_0q_7q_{11}q_{25}q_{28} - 2q_0q_9q_{10}q_{13}q_{32} + 2q_0q_9q_{11}q_{13}q_{29} \\ & - q_0q_9q_{11}q_{17}q_{26} + 2q_0q_9q_{11}q_{23}q_{26} - q_0q_9q_{11}q_{25}q_{26} + 2q_0q_9q_{13}q_{14}q_{27} \\ & - 2q_1q_7q_{11}q_{14}q_{28} + 2q_2q_3q_{11}q_{11}q_{33} - 2q_3q_9q_{11}q_{11}q_{26} + 2q_4q_7q_{10}q_{11}q_{28}\end{aligned}$$

HDQ-02: lower-theta quartic. Degree 4, multidegree $(3,0,2,1,1,1,2,1)$.

$$\begin{aligned}F_{HDQ-02} = {} & q_0q_{15}q_{16}q_{18} + q_0q_{16}q_{18}q_{23} - q_0q_{20}q_{22}q_{23} + q_1q_{14}q_{16}q_{18} \\ & - 2q_3q_{11}q_{16}q_{18} + q_3q_{11}q_{20}q_{22} - q_4q_{12}q_{18}q_{22} - q_8q_{10}q_{14}q_{16} \\ & + q_8q_{12}q_{14}q_{22}\end{aligned}$$

HDQ-03: lower-theta quartic. Degree 4, multidegree $(2,1,3,0,1,1,2,1)$.

$$\begin{aligned}F_{HDQ-03} = {} & q_0q_{16}q_{17}q_{29} - q_0q_{16}q_{23}q_{29} - q_0q_{17}q_{22}q_{31} + q_0q_{22}q_{23}q_{31} \\ & - q_1q_{14}q_{16}q_{29} + q_1q_{14}q_{22}q_{31} + 2q_3q_{11}q_{16}q_{29} - q_3q_{11}q_{22}q_{31} \\ & - q_3q_{14}q_{16}q_{27} - q_5q_{11}q_{22}q_{29} + q_5q_{14}q_{22}q_{27}\end{aligned}$$

HDQ-04: lower-theta quartic. Degree 4, multidegree $(2,1,3,0,2,1,1,1)$.

$$\begin{aligned}F_{HDQ-04} = {} & q_0q_{15}q_{16}q_{32} + q_0q_{16}q_{23}q_{32} - q_0q_{23}q_{24}q_{30} + q_1q_{14}q_{16}q_{32} \\ & - 2q_3q_{11}q_{16}q_{32} + q_3q_{11}q_{24}q_{30} - q_4q_{14}q_{16}q_{27} - q_6q_{10}q_{24}q_{32} \\ & + q_6q_{14}q_{24}q_{27}\end{aligned}$$

HDQ-05: theta3 cubic. Degree 3, multidegree $(1,1,1,1,1,1,1,1)$.

$$\begin{aligned}F_{HDQ-05} = {} & q_0q_{15}q_{35} + q_0q_{20}q_{30} + q_1q_{14}q_{35} + q_2q_{15}q_{33} \\ & + q_2q_{18}q_{32} - q_2q_{23}q_{33} - q_3q_{11}q_{35} - q_4q_{20}q_{26} \\ & + q_6q_{12}q_{33} - q_6q_{18}q_{28} - q_7q_{12}q_{32} - q_7q_{15}q_{28} \\ & + q_7q_{23}q_{28} - q_9q_{10}q_{30}\end{aligned}$$

Quintic port transports.

class	port permutation	source terms
25	$(0, 1, 3, 2)$	1599
26	$(0, 2, 1, 3)$	953
27	$(0, 2, 3, 1)$	1524
28	$(0, 3, 1, 2)$	985
29	$(0, 3, 2, 1)$	96
30	$(1, 0, 2, 3)$	730
31	$(1, 0, 3, 2)$	1988
32	$(1, 2, 0, 3)$	953
33	$(1, 2, 3, 0)$	2004
34	$(1, 3, 0, 2)$	985
35	$(1, 3, 2, 0)$	868
36	$(2, 0, 1, 3)$	730
37	$(2, 0, 3, 1)$	1722
39	$(2, 1, 3, 0)$	1807
40	$(2, 3, 0, 1)$	96
41	$(2, 3, 1, 0)$	868
42	$(3, 0, 1, 2)$	1988
43	$(3, 0, 2, 1)$	1722
44	$(3, 1, 0, 2)$	1599
45	$(3, 1, 2, 0)$	1807
46	$(3, 2, 0, 1)$	1524
47	$(3, 2, 1, 0)$	2004

Quartic coordinate transports.

source	class	base	coordinate action	port match
2	112	HDQ-02	identity	$(0, 1, 2, 3)$
2	113	HDQ-03	identity	$(0, 1, 3, 2)$
2	114	HDQ-03	identity	$(1, 0, 2, 3)$
2	115	HDQ-02	identity	$(1, 0, 3, 2)$
3	112	HDQ-02	identity	$(0, 1, 2, 3)$
3	113	HDQ-03	identity	$(0, 1, 3, 2)$
3	114	HDQ-03	identity	$(1, 0, 2, 3)$
3	115	HDQ-02	identity	$(1, 0, 3, 2)$
4	8	HDQ-04	identity	$(0, 1, 2, 3)$
4	9	HDQ-04	swap_01	$(0, 1, 3, 2)$
4	10	HDQ-04	swap_01	$(1, 0, 2, 3)$
4	11	HDQ-04	identity	$(1, 0, 3, 2)$

Cubic record bindings.

source	class	base	port match
5	9	HDQ-05	$(0, 1, 3, 2)$
5	10	HDQ-05	$(1, 0, 2, 3)$

8.4 Three worked paths

A direct quadratic separator. Template R4Q-03 is

$$q_{10}q_{20} - q_{12}q_{18}, \qquad \mathrm{mdeg} = (2,0,0,1,0,1,2,0).$$

It covers 252 canonical classes and 424 raw directions. For the representative `source_0:class_000000`, its source pullback has 44 nonzero terms and its target pullback is identically zero. The exact certificate is `d39c8b1f09360f48e9ae6e6ed552222d35cdc675fdd22c038781a7ff5e0328f5`, bound to the graph record by `10d7d0efe1d950f6f74585e2e385a5a07dfd33a2a51b56954a54b01c6b1aa44b`. This proves the stated source-to-target noncontainment; reversing the direction is not licensed.

A complete restoration root. For `s1:c658:t2792:p1032`, the following table prints every actual first- and second-child archetype. The row hash binds the role, insertion site, proof mechanism, transport fields, and evidence certificate.

layer	row	restored role	site	proof	status	row SHA-256
1	31976	`D_REPAIR_0_2`	0	`displayed_quartet_mismatch`	separated	`d2a1d79aaa37de82495c0db5478793b02dc15b7500f2783506614a3afc46531d`
1	31977	`D_REPAIR_0_2`	1	`displayed_quartet_mismatch`	separated	`18a009d352b311939bb21650fba5d66ce0a1a8dbe29ebaa7bd6393e0c0d25748`
1	31978	`D_REPAIR_0_2`	2	`full_map_Ti_zero_strict_sign`	separated	`a762631aa3a30f73764549778fcd2fceb9bcaac256430648a4f4ab1bdabb4869`
1	31979	`D_REPAIR_0_2`	3	`displayed_quartet_mismatch`	separated	`181aad684be16b6b531b5c2a8a8cbbd5ab4492018427e921140695868c4bc29e`
1	31980	`D_REPAIR_0_2`	4	`displayed_quartet_mismatch`	separated	`a37508b280187028982572253c3a59826137feb537228fb443fb79b684543626`
1	31981	`D_REPAIR_0_2`	5	`displayed_quartet_mismatch`	separated	`d7a4960b94bf787bdc5ddeafad029506a4693b5aa5876f5ecb947500abcf1a6c`
1	31982	`D_REPAIR_0_2`	6	`displayed_quartet_mismatch`	separated	`d5ad83a75f35c4b1bd795d89363c7b147d10ff6629930c327b3916883cd75ba2`
1	31983	`INCOMING`	0	`displayed_quartet_mismatch`	separated	`8246f706cb8e56b848238ca6e8715cc7ec257691cc03f567b97cac4843ec400a`
1	31984	`INCOMING`	1	`displayed_quartet_mismatch`	separated	`12bba2049bad09e6403af99ae1cd8f54003aafd14654396f2328220ac954b079`
1	31985	`INCOMING`	2	`displayed_quartet_mismatch`	separated	`8893a4e1763de61a221d08388243e53f69e501f2e89cee01d493e6ae6eb75f95`
1	31986	`INCOMING`	3	`displayed_quartet_mismatch`	separated	`07b72f4608262f6fc6958a0f24c966984c5d8d70c08f490abfa216cadb2c801e`
1	31987	`INCOMING`	4	`restore_remaining_physical_role`	continuation	`60636f150c4eb58cd64c0126bdb1d4aafe8e48927fc798974dfba76a5c51d272`
1	31988	`INCOMING`	5	`displayed_quartet_mismatch`	separated	`718c7d8d3fd230a2bd1b25ecbb0faff759a6bb6a8b0cd261f530806b3aac25b7`
1	31989	`INCOMING`	6	`displayed_quartet_mismatch`	separated	`b86c1e7ab3211f299a5b9eb5910e3ef3b3ae35f69065cf189be6e31d04d288b8`
2	1	`D_REPAIR_0_2`	0	`displayed_quartet_mismatch`	separated	`daddc54b748d86b7f3698b771e1ea090b3fee641becc9b68fb76b0775d2b48e1`

layer	row	restored role	site	proof	status	row SHA-256
2	2	D_REPAIR_0_2	1	displayed_quartet_mismatch	separated	3d678ac9082f6b02ffebb34085488455455345b13f1e037681b75051c81a603e
2	3	D_REPAIR_0_2	2	full_map_Ti_zero_strict_sign	separated	dbe28657406837bfe91de37dc42fb124c9a71d2c475bf1655811deeeac55ed00
2	4	D_REPAIR_0_2	3	displayed_quartet_mismatch	separated	e655cf2cf0aa1572b4a83ad7b24bbefde08e7fd968b0906c31cc3beb6977d72a
2	5	D_REPAIR_0_2	4	displayed_quartet_mismatch	separated	f61fd9ab276190e4e08a3e91288d012d46fad0e92ef586e74c9c20da566280d2
2	6	D_REPAIR_0_2	5	displayed_quartet_mismatch	separated	5145c0082ec9fc0bfc40d4315a851cf39f634e4ea3b14504f3f455a8139b0233
2	7	D_REPAIR_0_2	6	displayed_quartet_mismatch	separated	61bd0a2485ca2ba94809531aae4245dd53821a10a041f95ff1e33b1f20f118a1
2	8	D_REPAIR_0_2	7	displayed_quartet_mismatch	separated	1b825df66c3c7d4f9c9cf2156fa16cc7dd1fe9dc0849deb83d0203bfcea3a64d

There are 14 first children: 12 displayed-quartet terminals, one full-map $\mathcal{T}_i$ terminal, and one continuation (row 31987). That continuation has exactly eight children: seven displayed-quartet terminals and one full-map $\mathcal{T}_i$ terminal. Every second child has an empty remaining-role list. Hence this exact subtree has 21 leaves, depth 2, zero unresolved records, and no cycle.

A one-/two-port word transport. The anchor `tree:k3:identity` is the labelled identity tree transport `4342a65ef53537888b371724949a9ec3260731896777270ececd8fea0b475a7f`. At source and target site zero, inserting label 3 gives the exact one-port transport `c2ed6cbed9d725664268416f315268cfdab6e2c2844790c7596aab7010e0a4e0`; both parent restrictions are `R:57a56588cb2934a8ad6d9500e0d28a8f7a3e8f797ba05e6a213af02a676eb539`. Inserting label 4 at the new site zero gives the exact two-port transport `8e3c37f3b5c32225fbf97286be01da6754a33e4fd267e4741a918740d5a4c8ba`, whose parent transport is precisely the one-port transport. Its two restriction records are `R:b286c2133804e765ea7889cbdc790218c360023201fbd93991e608af071e4bd3` and `R:0bf3e2b848168b2d5c1d689601f1c1700bd3f98502524f32763d20ddac093639`. The reversed marginal is the exact one-port transport `c144b9418ca2bc530028a613a74290e7ded74d9b38589f323cd7e01b7bdf33b6` on the same base anchor. The selected ledger rows have hashes `67691e6e518436f8911fe0dc82402de996b0deef87e3f7ad6396fb0b81bd0ef0` and `44c2b033d79916a3226985efc6211b4b54844953f90a7c00b2788a0cfe206193`. Thus both deletion orders are bound, not inferred from an unrecorded symmetry.

8.5 Provenance and replay

field	exact value
schema/version	`k2p-printed-certificate-appendix-v1`
producer	`.venv/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -write`
deterministic check	`.venv/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -check`
independent replay	`.venv/bin/python -B proof_compression_submission/templates/verify_printed_certificate_appendix.py`
mutation suite	`.venv/bin/python -B proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py`
sealed payload	`ea588171aa43f6bac78b1047e8e9b5a40228eceafb595f52608fab29d16e4237`

The authoritative input bindings are:

path	SHA-256
`proof_compression_submission/templates/DIRECT_CERTIFICATE_TEMPLATE_TABLE.json`	`1617f1ea3287cd5a9030617e7a1bb9f47f29b4508197871c77fbbd22ba2f3e4f`
`work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz`	`0a1818655429d60660c1ed87f3fbe412701f386b081562b3a4caa54079069f1d`
`package/referee/k2p_offline_sweep_portable/proofs/theta0_quintic_orbit_certificate.json`	`f863afd5875a74be818141990863344fae09fd4269a803d3a1bfeb67a8a595e0`
`package/referee/k2p_offline_sweep_portable/proofs/theta_quartic_obstruction_certificates.json`	`1127004383145ae1fa053a86e43199e3f627fefe8cd978feefc80b9b89e7b696`
`package/referee/k2p_offline_sweep_portable/proofs/theta3_cubic_obstruction_certificate.json`	`d1501a7e86b6b2b614590454ba4a44afc04785cb19711b70d2cd8069bd35e0bf`
`package/referee/k2p_offline_sweep_portable/proofs/four_port_direct_residual_closure_certificate.json`	`9dc3f112d0b4ed8883160b5c53111cd5208f911546cbb63d1b99a76d7f53f861`
`work/restoration_sign_reclassification/corrected_restoration_forest.json`	`bcf91bf433c71056d1e27871dd15fe532f9ae1cc4ad79eb2373eae57071ee427`
`work/probe_coherence_corrected/probe_coherence_certificate.json`	`53d1ac1e6a14637f547c031a6e8031d3d3cc49630518f31339813031089e0bfc`
`work/probe_coherence_corrected/one_port_ledger.jsonl.gz`	`d5fa13d38731bff2403eeb4e4d9e139566c4983b09d30553c6260eaac64c5c90`
`work/probe_coherence_corrected/two_port_ledger.jsonl.gz`	`10f0afcab77f2d61cecfc36d723c6f32065c304ac088b0b8ecf12dfc867fbf9d`
`work/probe_coherence_corrected/parent_restriction_ledger.jsonl.gz`	`5d1e6c2fe38d31f6304a76886ec37829215b88c8b179f5b23596d49d37ceeb38`
`work/probe_coherence_corrected/exact_transport_ledger.jsonl.gz`	`6bc8e88feac2bee68491287775f078e8e5474bf930961a7390967c9fd350044d`

9 Full weak-class certificate

The base graphs are encoded by

$$\begin{aligned} W: &\quad rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2, &\qquad \mathrm{Ret} = \{V, X\}, \\ W': &\quad rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2, &\qquad \mathrm{Ret} = \{X_0, X_1\}. \end{aligned}$$

Complete rooting censuses are $(5, 2, 3)$ and $(7, 2, 5)$ for admissible, tree-child, and non-tree-child rootings. Exact mixed-incidence comparison returns relation `none`, including after forgetting head flags on the unique triangle of each graph.

Put $\delta = 2^{-30}$. The complete parameter mapping is:

datum	W	W'
seven internal pairs	$(1/7, 1/7)$	$(1/4, 1/4)$
inheritance 1	$Z \to X : 15996/16339;\ U \to X : 343/16339$	$V \to X_1 : 1/2;\ U \to X_1 : 1/2$
inheritance 2	$S \to V : 1/8;\ U \to V : 7/8$	$V \to X_0 : 1/6;\ S \to X_0 : 5/6$
leaf 0 pair	$(86779\delta/80, 86779\delta/80)$	$(16\delta/3, 16\delta/3)$
leaf 1 pair	$(320\delta/253, 320\delta/253)$	$(32\delta/9, 32\delta/9)$
leaf 2 pair	$(114373\delta/20240, 114373\delta/20240)$	$(96\delta/5, 96\delta/5)$

In character code order $0, C, G, T = 0, 1, 2, 3$, the common coordinate order is

$$(000, 0CC, 0GG, C0C, CC0, CGT, CTG, G0G, GCT, GG0).$$

The value is 1 at 000, δ^2 at the six two-nonzero coordinates, and $4\delta^3/5$ at the three all-nonzero coordinates.

The normalized tensors before pendant scaling are, respectively,

$$\left(1, \frac{64009}{457492}, \frac{64009}{457492}, \frac{6400}{39229939}, \frac{1}{1372}, \frac{4048}{39229939}, \frac{4048}{39229939}, \frac{6400}{39229939}, \frac{4048}{39229939}, \frac{1}{1372}\right)$$

and

$$\left(1, \frac{15}{1024}, \frac{15}{1024}, \frac{5}{512}, \frac{27}{512}, \frac{9}{4096}, \frac{9}{4096}, \frac{5}{512}, \frac{9}{4096}, \frac{27}{512}\right).$$

The graph-derived canonical orders of the visible edge classes are

$$W : (ZX, SV, rS, SU, UV, VZ, UX), \qquad W' : (VX_1, VX_0, UV, rS, SX_0, SU, UX_1).$$

Accordingly, columns $0, \ldots, 8$ mean

$$W :;(s_{ZX}, g_{ZX}, s_{SV}, g_{SV}, s_{rS}, g_{rS}, s_{SU}, g_{SU}, s_{UV}),$$
$$W' :;(s_{VX_1}, g_{VX_1}, s_{VX_0}, g_{VX_0}, s_{UV}, g_{UV}, s_{rS}, g_{rS}, s_{SX_0}).$$

These names are a derived crosswalk, not an unstated field of the frozen certificate: the crosswalk verifier reconstructs the descendant signatures from the two primitive graphs, binds the frozen graph and descriptor hashes, and rejects a reordered column. The sealed artifact and independent checker are, respectively,

`proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json`

`proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py`.

All output-row indices in the following table are zero-based in the displayed Fourier-coordinate order; row 0 is the all-zero coordinate. The minors' rows and determinants are

map	output rows	determinant
W	$(1, 2, 3, 5, 4, 7, 6, 8, 9)$	10368019213741323/563981315074464023964442388464888915634290688
W'	$(1, 2, 3, 5, 4, 6, 7, 8, 9)$	1435825/85002596691653613846528

For cherry induction use actual physical pairs $(u_s, u_g) = (2/5, 4/9)$ and $(v_s, v_g) = (3/7, 5/11)$. The observable block $(u_s/v_s, u_s v_s, u_g/v_g, u_g v_g)$ has determinant 2464/675. More explicitly, for an outside leaf j and new cherry leaves a, b,

$$\frac{Q_{j=C,a=C,b=0}}{Q_{j=C,a=0,b=C}} = \frac{u_s}{v_s}, \qquad Q_{a=C,b=C} = u_s v_s,$$

with all omitted characters zero; replacing C by G gives the other two observables. Square-root recovery gives the four edge factors, and division by the corresponding nonzero factor recovers every old tensor coordinate. This is the analytic local inverse; graph pruning separately preserves weak-not-strong status and inequivalence.

10 Theorem-to-artifact crosswalk

All paths below are relative to the `k2p_level2_identifiability_closure` project root. "Primary" is the producer/certificate path; "replay" is a separately implemented checker or audit. Historical files not named here are not promotion evidence. This printed table is a reader map, not the complete machine schema. The field-rich authority is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`, which records each claim's declared artifact schemas, exact byte hashes, producer, replay and mutation artifacts, environment profile, and whether a runtime is known or deliberately null. Executable entry points are invoked from the project root with the pinned Python environment shown in Section 12.

claim	primary proof/certificate	independent replay or mutation
K2P domain, subdivision, rooting	`work/domain_rooting_closure/PROOF.md`; `work/domain_rooting_closure/domain_rooting_certificate.json`	`work/domain_rooting_closure/verify_domain_rooting.py`
Quartet signs and tree of blobs	`work/quartet_separation_closure/PROOF.md`; `work/quartet_separation_closure/quartet_logic_certificate.json`	`work/quartet_separation_closure/verify_quartet_logic.py`; `work/adversarial_proof_review/verify_topology_direction.py`
Whole-map $\mathcal{T}_i$ and triangle germ	`package/original/checkpoint_2/continuation_2/K2P_TREE_SUNLET_SIGN_CERTIFICATE.md`; `package/original/checkpoint_2/continuation_2/K2P_TRIANGLE_GERM_EXACT.md`	`package/original/checkpoint_2/continuation_2/verify_triangle_and_sunlet.py`; `work/final_theorem_release/no_assert_triangle_sunlet.py`
Bridge fibre, marginal, gluing	`work/bridge_marginal_closure/PROOF.md`; `work/bridge_marginal_closure/certificate.json`	`work/bridge_marginal_closure/verify_bridge_marginal.py`; `work/adversarial_proof_review/PHYSICAL_LOCAL_PRODUCT_REPAIR.md`; `work/adversarial_proof_review/verify_adversarial.py`
Global classification, genericity, reconstruction, and continuous time	`proof_compression_submission/article/main.tex`; `work/global_theorem_closure/promotion_manuscript/K2P_SAME_PROMOTION_MANUSCRIPT.md`	`work/global_theorem_closure/promotion_manuscript/QUANTIFIER_AUDIT.md`; `work/global_proof_adversary/AUDIT.md`
Four-port raw universe and terminals	`work/corrected_composite_ledgers/artifacts/raw4_corrected_composite_ledger.jsonl.gz`; `work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz`	`work/corrected_composite_ledgers/verify_corrected_composites_independent.py`; `work/corrected_composite_ledgers/run_composite_mutations.py`
Five-port θ_2	`work/corrected_composite_ledgers/artifacts/theta2_corrected_composite_ledger.jsonl.gz`	`work/corrected_composite_ledgers/verify_corrected_composites_independent.py`; `work/corrected_composite_ledgers/run_composite_mutations.py`, family `theta2`

claim	primary proof/certificate	independent replay or mutation
Production direct residual closure	package/referee/k2p_offline_sweep_portable/proofs/four_port_direct_residual_closure_certificate.json; package/referee/k2p_offline_sweep_portable/DIRECT_CLOSURE_LOCK.json	package/referee/k2p_offline_sweep_portable/verify_direct_closure_release.py; package/referee/k2p_offline_sweep_portable/test_direct_closure_release_mutations.py; package/referee/k2p_offline_sweep_portable/direct_closure_mutation_report.json
Printed direct templates and worked paths	proof_compression_submission/templates/PRINTED_CERTIFICATE_APPENDIX.json; proof_compression_submission/supplement/certificate_appendix.tex	proof_compression_submission/templates/verify_printed_certificate_appendix.py; proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py (reader-facing presentation regression only; theorem authority remains the production direct-closure replay and mutation suite)
997-parent forest	work/restoration_sign_reclassification/corrected_restoration_forest.json	work/restoration_sign_reclassification/corrected_restoration_replay_certificate.json; work/restoration_sign_reclassification/verify_corrected_restoration_forest.py; work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py
Three-port cycle closure	work/cycle_three_port_closure/PROOF.md; work/cycle_three_port_closure/promotion/cycle_promotion_certificate.json	work/adversarial_proof_review/verify_corrected_cycle_promotion.py; work/adversarial_proof_review/test_cycle_promotion_mutations.py
Probe input completeness	work/adversarial_proof_review/PROBE_INPUT_CONTRACT.md; work/adversarial_proof_review/probe_input_contract.json	work/adversarial_proof_review/verify_probe_input_contract.py; work/adversarial_proof_review/test_probe_input_mutations.py

claim	primary proof/certificate	independent replay or mutation
One-/two-port probe closure	work/probe_coherence_corrected/PROOF.md; work/probe_coherence_corrected/probe_coherence_certificate.json	work/probe_coherence_corrected/probe_coherence_independent_verification.json; work/probe_coherence_corrected/verify_probe_coherence_corrected.py; work/probe_coherence_corrected/run_probe_coherence_mutations.py
Primitive probe audit	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit_certificate.json	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit.py; work/global_proof_adversary/probe_full_audit/independent_probe_mutation_report.json
Weak sharpness	work/weak_sharpness_closure/PROOF.md; work/weak_sharpness_closure/weak_sharpness_certificate.json; proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json	work/weak_sharpness_audit/PROOF_AUDIT.md; work/weak_sharpness_audit/audit_certificate.json; work/weak_sharpness_audit/test_mutations.py; work/weak_sharpness_audit/mutation_report.json; proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py; proof_compression_submission/analysis/test_weak_sharpness_column_crosswalk_mutations.py
Unified theorem lock	work/final_theorem_release/RELEASE_LOCK.json; work/final_theorem_release/verify_final_theorem_release.py	work/final_theorem_release/corrected_universe_independent_replay.json; work/final_theorem_release/run_release_mutations.py

10.1 Printed finite-layer replay metadata

For the principal computed layers, each row below supplies the requested schema/version, authoritative file SHA-256, producer command, independent replay command, and mutation command. Commands run from the project root; none may be invoked with optimized Python.

layer	exact machine metadata
raw four-port topology ledger	schema/version: `k2p-four-port-raw-ledger-summary-v1`; authority: `work/raw_ledger_audit/artifacts/raw_ledger_summary.json`; SHA-256: `82cab09acb54ac0caac7721acebc84285fef8a2ae166e1808c31e9bdad1f7a79`; producer command: `.venv/bin/python -B work/raw_ledger_audit/generate_raw_ledger.py`; replay command: `.venv/bin/python -B work/raw_ledger_audit/verify_raw_ledger.py`; historical mutation-regression command (retained for provenance, not current theorem qualification): `.venv/bin/python -B work/raw_ledger_audit/test_mutations.py`. The current semantic rank attack is the exact-rank production-verifier suite in the next row.
four-port exact-rank filter	schema/version: `k2p-four-port-exact-generic-rank-upper-coverage-v1`; authority: `work/rank_upper_certificates/rank_upper_coverage.json`; SHA-256: `c52c5730494eb894360c17b6e54ae5c260fca3cddb8702d5c796750c7df874bc`; producer command: `PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/build_rank_upper_coverage.py`; replay command: `PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/verify_rank_upper_certificates.py`; mutation command: `PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/mutation_tests.py -output /tmp/k2p-rank-upper-mutations.json`.
direct terminal templates	schema/version: `k2p-printed-certificate-appendix-v1`; reader derivative: `proof_compression_submission/templates/PRINTED_CERTIFICATE_APPENDIX.json`; SHA-256: `facfeb75b1a15958f0e665479b3aa2bf4c8b011c5a79622910ef8bf9f52a9900`; producer command: `.venv/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -write`; replay command: `.venv/bin/python -B proof_compression_submission/templates/verify_printed_certificate_appendix.py`; presentation-regression command (not theorem qualification): `.venv/bin/python -B proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py`.
997-parent restoration forest	schema/version: `k2p-corrected-restoration-forest-v3`; authority: `work/restoration_sign_reclassification/corrected_restoration_forest.json`; SHA-256: `bcf91bf433c71056d1e27871dd15fe532f9ae1cc4ad79eb2373eae57071ee427`; producer command: `.venv/bin/python -B work/restoration_sign_reclassification/build_corrected_restoration_forest.py`; replay command: `.venv/bin/python -B work/restoration_sign_reclassification/verify_corrected_restoration_forest.py`; mutation command: `.venv/bin/python -B work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py -output /tmp/k2p-restoration-mutations.json`.
one-/two-port probe closure	schema/version: `k2p-corrected-coherent-probe-closure-v1`; authority: `work/probe_coherence_corrected/probe_coherence_certificate.json`; SHA-256: `53d1ac1e6a14637f547c031a6e8031d3d3cc49630518f31339813031089e0bfc`; producer command: `.venv/bin/python -B work/probe_coherence_corrected/build_probe_coherence_corrected.py`; replay command: `.venv/bin/python -B work/probe_coherence_corrected/verify_probe_coherence_corrected.py`; mutation command: `.venv/bin/python -B work/probe_coherence_corrected/run_probe_coherence_mutations.py -output /tmp/k2p-probe-mutations.json`.
weak-sharpness named columns	schema/version: `k2p-weak-sharpness-column-crosswalk-v1`; authority: `proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json`; SHA-256: `c3b302c37744fa33834c75ca5b22da9d374d5def4f1e6f4650e3b2b3b4166437`; producer command: `.venv/bin/python -B proof_compression_submission/analysis/build_weak_sharpness_column_crosswalk.py`; replay command: `.venv/bin/python -B proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py`; mutation command: `.venv/bin/python -B proof_compression_submission/analysis/test_weak_sharpness_column_crosswalk_mutations.py`.
full-map certificate reseal	schema/version: `k2p-full-map-domain-reseal-audit-v1`; authority: `work/final_theorem_release/full_map_reseal_audit.json`; SHA-256: `945c957f0b73ab01d9c5bd6942ccb881f86e22fc47d8e4cc1fff5da8f4a64f40`; producer/replay/mutation command: `.venv/bin/python -B work/final_theorem_release/verify_full_map_reseal.py -output /tmp/k2p-full-map-reseal-audit.json`.
composite reseal differential	schema/version: `k2p-composite-domain-reseal-diff-audit-v1`; authority: `work/final_theorem_release/composite_reseal_diff_audit.json`; SHA-256: `bc91fee3b7541fcae72c4db2e66776fbfc69c43890718239f0eea41bb2cc0654`; producer/replay command: `.venv/bin/python -B work/final_theorem_release/verify_composite_reseal_diff.py`; mutation authority: the six fail-closed seal/domain mutations in the preceding reseal audit.

11 Frozen hash anchors

The following SHA-256 values are the compact anchors most useful for checking that the intended finite universe has been unpacked. The outer release lock authenticates all nested files and should remain the authority if any table is updated.

artifact	file SHA-256
unified release lock	`130642e235c9beaa22061c578c3c645244cdbf45a9b416d45d94492b3d2848bd`
raw-four composite ledger	`c6cd9d6b5b09371565fd3e58ff9ab3cd7266b6231b153d43f9d1e886af8eae27`
raw-four independent replay	`1a4ac5c5ab5f86228f9e59c62a9021547907a9d6238e1171a7074f49506a8c66`
raw-four terminal registry	`d7b52d28f3d85db8f63bafa984f6341d8e11487298e7d3c9216edffe586306fa`
raw-four mutation report	`db6d4e6c8986db20ca623724981d2d4f39f6ff0ccf5d70e708190c1e09a86d4a`
five-port θ_2 composite ledger	`805fc7f5a3de9dad2c63a210208075cf19910cf811ffd08878f32782ce71b659`
five-port independent replay	`dd02a752b2ce41628039d2f6da6fdab77f2a8ffd73b8cd80e2968790dbbf3150`
five-port mutation report	`ec2c6ec092539048b4e7ab9d9cfea01caa985d0f35cae74ca56732dc4cfe4c84`
full-map certificate reseal audit	`945c957f0b73ab01d9c5bd6942ccb881f86e22fc47d8e4cc1fff5da8f4a64f40`
composite reseal differential audit	`bc91fee3b7541fcae72c4db2e66776fbfc69c43890718239f0eea41bb2cc0654`
corrected 997-parent forest	`bcf91bf433c71056d1e27871dd15fe532f9ae1cc4ad79eb2373eae57071ee427`
restoration independent replay	`42be6b0c4d85aa58b336caebbdefd10a0af0ce4234a0482e65c7b5a68d1e6430`
probe input contract	`71d8596c5e0fa5804f5a1d938423ba9802f4d11783ef0e9d0f45a0453f0aff22`
full probe primary	`53d1ac1e6a14637f547c031a6e8031d3d3cc49630518f31339813031089e0bfc`
full probe independent graph audit	`74cd1dfb4d6c38dfe73b8bb3c76d29b22eb5f8ab8ca80a9d56bc376ec5d0fa0d`
full probe adversarial mutation report	`70fd78df07220d7ab27ba477ed584f3ba1c5cc45a96cd4d4ffdb5d1a34b1325a`
weak-sharpness primary	`e66c78a0aeab990b4dc448f4f064b37e1e15ecbff75a5f472bf116d4464378bd`
weak-sharpness independent audit	`cfd8d3a2ebc7431d141cac6ebe943e25730eb086fbc84b52833a40bee40a5d52`

The combined one-/two-port ordered-ledger root is

`7868fed6f8e0c10fcb9740da8ffdcb7f64ea68939c99cba6f364da4cfd90bf50`.

12 Replay protocol

Run from the project root. The locked environment uses Python 3.10 or newer; the qualified release used Python 3.14.6, NetworkX 3.5, and SymPy 1.14.0.

```
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install \
  -r work/final_theorem_release/requirements.txt

.venv/bin/python -B \
  work/final_theorem_release/build_release_lock.py --check --require-ready
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --quick
.venv/bin/python -B \
  work/final_theorem_release/run_release_mutations.py
```

For full primitive regeneration:

```
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --full
```

The entry points reject optimized Python (`python -O`). Full mode regenerates large ledgers by streaming; it does not require an uncompressed multi-gigabyte temporary JSONL file. The submission bundle requires a detached clean-checkout full replay of the exact candidate sources with every layer passing, zero blockers, and `promotion_ready=true`. Its exact candidate commit, layer count, internal and wall times, memory measurements, report SHA-256, and telemetry SHA-256 are deliberately not compiled into this source. They are recorded in `proof_compression_submission/output/FINAL_CLEAN_FULL_REPLAY.json`, `proof_compression_submission/output/FINAL_CLEAN_FULL_REPLAY_TELEMETRY.json`, and the generated theorem-to-artifact crosswalk. This separation lets the replay byte-bind these exact submission sources without a self-referential source edit after the run.

The frozen five-port θ_2 artifacts retain the compiler, canonicalizer, and input-lock hashes of their original generation. The full replay requires those legacy bindings exactly and reconstructs the current expected bytes by changing only those provenance fields, the two consequent artifact-metadata rows, and the summary seal. It then compares the freshly generated rank and restoration payloads with those reconstructed bytes and requires the direct-proof, class-partition, and all 2,946,240 raw-ledger rows to remain literally byte-identical. Dedicated mutations reject a wrong legacy binding; the outer release lock independently fixes the current compiler, canonicalizer, and input lock.

13 Mutation coverage

The independent suites reject at least the following semantic faults:

- omitted or duplicated raw relation;
- swapped or false topology reason;
- false target rank upper bound or rank exclusion;
- reassignment of a terminal class;
- omitted restoration role or child;
- wrong source insertion, target attachment, or parent;
- restoration cycle or missing continuation layer;
- broken boundary transport or parent restriction;
- omitted probe anchor, pendant site, reticulation-incoming site, or root-suppressed site;
- splitting the two artificial root halves into distinct physical sites;
- source–target reversal omission;
- reassignment of quadratic, cubic, quartic, quintic, or $\mathcal{T}_i$ certificate;
- reintroduction of a rooted `triple_type` topology oracle;
- changed sharpness arrows, parents, inheritance complements, Fourier coordinates, rank minors, or incomplete rooting list; and
- optimized-Python execution that would disable fail-closed checks.

Mutation rejection demonstrates that the verifier watches these data. It is not a substitute for the direct mathematical implication associated with each certificate type.

14 Recent-source verification note

Citation metadata were checked on 21 August 2026 against primary publisher or authoritative preprint records where available:

source	verified locator and status
Kimura (1980) [10]	Springer version of record, https://doi.org/10.1007/BF01731581, vol. 16, pp. 111–120.
Evans–Speed (1993) [5]	Project Euclid/Annals record, https://doi.org/10.1214/aos/1176349030, vol. 21, pp. 355–377.
Sturmfels–Sullivant (2005) [12]	author/arXiv and journal metadata agree on J. Comput. Biol. 12(2), pp. 204–228, https://doi.org/10.1089/cmb.2005.12.204.
Gross et al. (2021) [6]	Springer version of record, https://doi.org/10.1007/s00285-021-01653-8, J. Math. Biol. 83:32.
Cox–Gross–Martin (2025) [3]	Springer version of record, https://doi.org/10.1007/s11538-025-01506-1, Bull. Math. Biol. 87:132.
Ardiyansyah (2021) [1]	arXiv primary record https://arxiv.org/abs/2104.12479v1; the abstract states the simple/semisimple level-2 invariant scope used in the related-work summary.
Holtgrefe et al. (2025) [7]	Springer version of record, https://doi.org/10.1007/s11538-025-01549-4, Bull. Math. Biol. 87:168; the publisher records a 14 January 2026 correction.
Holtgrefe et al. (2026) [8]	Springer version of record, https://doi.org/10.1007/s12064-025-00453-8, Theory Biosci. 145:4.
Huber et al. (2025) [9]	Springer version of record, https://doi.org/10.1007/s11538-025-01510-5, Bull. Math. Biol. 87:136; Lemma 4.2 is the two-generator result used here.
Englander et al. (2025) [4]	bioRxiv DOI https://doi.org/10.1101/2025.04.18.649493; theorem numbering and the 4 July 2026 version-4 content were checked against the locally archived source XML/PDF.
Brits et al. (2026) [2]	arXiv primary record https://arxiv.org/abs/2607.12919v3; version 3 dated 25 August 2026. The level-1 JC/K2P/K3P result used here remains Theorem 4.9; the version-3 narrowing of the separate arbitrary-level tree–network statement to JC is not used.
Kriebel (2026) [11]	versioned public companion JC repository release https://github.com/AlecKriebel/Math/releases/tag/stc-jc-sharp-boundary-v1.1.4, source commit `b5b855f5e6552e1d87f32d90851ea2def5330364`; no DOI is claimed.

The local K2P/K3P theta-collision manuscript is provisional and has no verified public bibliographic status in this draft. It is therefore not cited as a published result or used as theorem evidence.

15 Submission metadata and licenses

The corresponding address is me@aleckriebel.com. No specific funding supported this work, and the author declares no competing interests. The article, supplement, and certificate data are licensed under CC BY 4.0; verifier code is licensed under the MIT License. The public repository is https://github.com/AlecKriebel/Math, the project path is `k2p_level2_identifiability_closure`, and the immutable source tag for this submission is `k2p-same-biorxiv-v1.0.2`. The final deterministic archive SHA-256 is recorded alongside the submission package. No GitHub Release, Zenodo deposit, or DOI is claimed in this version.

Disclosure

Generative AI assisted with discovery, code, drafting, and adversarial review. The cited independent replays are separately implemented programs, not independent human review. The author accepts responsibility for the final mathematics, software, and submission.

References

[1] Muhammad Ardiyansyah. Distinguishing level-2 phylogenetic networks using phylogenetic invariants, 2021.

[2] Jari Brits, Niels Holtgrefe, Leo van Iersel, and Samuel Martin. On tree–network distinguishability and full identifiability of phylogenetic networks, 2026. Version 3, 25 August 2026; Theorem 4.9.

[3] Shelby Cox, Elizabeth Gross, and Samuel Martin. Group-based phylogenetic models on 3-sunlet networks. *Bulletin of Mathematical Biology*, 87:132, 2025.

[4] Aviva K. Englander, Martin Frohn, Elizabeth Gross, Niels Holtgrefe, Leo van Iersel, Mark Jones, and Seth Sullivant. Identifiability of phylogenetic level-2 networks under the jukes–cantor model. *bioRxiv*, 2025. bioRxiv preprint, first posted 24 April 2025; version 4 revised 4 July 2026.

[5] Steven N. Evans and Terry P. Speed. Invariants of some probability models used in phylogenetic inference. *The Annals of Statistics*, 21(1):355–377, 1993.

[6] Elizabeth Gross, Leo van Iersel, Remie Janssen, Mark Jones, Colby Long, and Yukihiro Murakami. Distinguishing level-1 phylogenetic networks on the basis of data generated by markov processes. *Journal of Mathematical Biology*, 83(3):32, 2021.

[7] Niels Holtgrefe, Elizabeth S. Allman, Hector Baños, Leo van Iersel, Vincent Moulton, John A. Rhodes, and Kristina Wicke. Distinguishing phylogenetic level-2 networks with quartets and inter-taxon quartet distances. *Bulletin of Mathematical Biology*, 87(12):168, 2025.

[8] Niels Holtgrefe, Katharina T. Huber, Leo van Iersel, Mark Jones, and Vincent Moulton. Characterizing semi-directed phylogenetic networks and their multi-rootable variants. *Theory in Biosciences*, 145:4, 2026.

[9] Katharina T. Huber, Leo van Iersel, Mark Jones, Vincent Moulton, and Leonie Veenema-Nipius. When are quarnets sufficient to reconstruct semi-directed phylogenetic networks? *Bulletin of Mathematical Biology*, 87(10):136, 2025.

[10] Motoo Kimura. A simple method for estimating evolutionary rates of base substitutions through comparative studies of nucleotide sequences. *Journal of Molecular Evolution*, 16(2):111–120, 1980.

[11] Alec Kriebel. Strong tree-childness is a sharp generic-identifiability boundary for level-2 jukes–cantor networks. GitHub release, 2026. Companion manuscript and reproducibility package; source commit b5b855f5e6552e1d87f32d90851ea2def5330364.

[12] Bernd Sturmfels and Seth Sullivant. Toric ideals of phylogenetic invariants. *Journal of Computational Biology*, 12(2):204–228, 2005.